

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3

Total Marks:30

Annexure C

Case study

Criteria	Understandin g of Case (2.5)	Application of Concepts (2.5)	Analysi s (2)	Solution Design (2)	Clarity (1)	Total(10)
Weighttag e	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						

Code of Conduct:

I S.Vyshnavi(192472041)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Vyshnavi

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

Question 1: Shepherd Boy Case – Confusion Matrix

A shepherd boy repeatedly tells villagers that a **wolf is coming**, even when there is no wolf.

Later, a real wolf comes, but the villagers don't believe him.

The question is asking us to identify **TP, FP, FN and TN**.

First understand the terms

Here:

- **Positive = Wolf is present**
- **Negative = Wolf is not present**
- **Prediction = Shepherd's warning**

1. True Positive (TP)

A wolf is **actually present**, and the shepherd **says "Wolf!"**

So both the actual condition and prediction are positive.

TP = 1

Correctly detected a wolf.

2. False Positive (FP)

There is **no wolf**, but the shepherd **says "Wolf!"**

The prediction is positive, but the actual condition is negative.

FP = 1

This is a **false alarm**.

3. False Negative (FN)

A wolf is **actually present**, but the shepherd **doesn't say "Wolf!"**

The actual condition is positive, but the prediction is negative.

The story does not explicitly give this situation.

FN = 0 (observed in the given story)

4. True Negative (TN)

There is **no wolf**, and the shepherd **doesn't say "Wolf!"**

Both actual and prediction are negative.

Again, this situation is not explicitly given in the story.

TN = 0 (observed in the given story)

Final answer:

	Actual Wolf	Actual No Wolf
Says Wolf	TP = 1	FP = 1
Doesn't Say Wolf	FN = 0	TN = 0

Main idea

The story demonstrates how a prediction can be:

Correct positive → TP

Wrong positive → FP

Wrong negative → FN

Correct negative → TN

Question 2: Cat Image Classification

This question gives us **100 images**.

Given information:

- **30 images actually contain cats**
- **70 images do not contain cats**
- **Model correctly identifies 25 cat images**
- **Model incorrectly predicts 50 non-cat images as cats**

We need to calculate:

TP, FP, FN, TN, Precision, Recall and F1 Score.

Step 1: Find TP

The model correctly identifies **25 cat images**.

Actual = Cat

Prediction = Cat

Therefore:

$$\mathbf{TP = 25}$$

These are correctly classified cat images.

Step 2: Find FN

There are **30 actual cat images**.

The model correctly identified only **25**.

So the remaining cat images were missed:

$$\text{FN} = 30 - 25$$

$$\text{FN} = 5$$

These are actually cats, but the model says they are not cats.

Step 3: Find FP

The question says the model predicts **50 of the non-cat images as cats**.

Actual = Not Cat

Prediction = Cat

Therefore:

$$\text{FP} = 50$$

These are false alarms.

Step 4: Find TN

There are **70 non-cat images**.

Out of these, 50 were incorrectly predicted as cats.

So the remaining 20 were correctly identified as non-cat:

$$\text{TN} = 70 - 50$$

$$\text{TN} = 20$$

Confusion Matrix

Now we have:

	Actual Cat	Actual Not Cat
Predicted Cat	TP = 25	FP = 50
Predicted Not Cat	FN = 5	TN = 20

Check total:

$$25 + 50 + 5 + 20 = 100$$

So everything is correct.

Precision

Precision tells us:

When the model predicts "Cat", how often is it actually a cat?

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Substitute:

$$= 25 / (25 + 50)$$

$$= 25 / 75$$

$$= 0.3333$$

Therefore:

$$\text{Precision} = 33.33\%$$

This is relatively low because the model incorrectly predicted **50 non-cat images as cats**.

Recall

Recall tells us:

Out of all the actual cat images, how many did the model correctly identify?

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$= 25 / (25 + 5)$$

$$= 25 / 30$$

$$= 0.8333$$

Therefore:

$$\text{Recall} = 83.33\%$$

This is high because the model found **25 out of 30 actual cats**.

F1 Score

F1 Score combines **Precision and Recall**.

Formula:

$$\text{F1} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

Using:

- Precision = 0.3333
- Recall = 0.8333

We get:

$$\text{F1} \approx 47.62\%$$

So:

Precision = 33.33%

Recall = 83.33%

F1 Score = 47.62%

Question 3: Naïve Bayes – Spam Email

This question is about using **Naïve Bayes classification** to decide whether a new email is:

Spam or Not Spam

The dataset contains 6 emails.

Offer Free Length Class

Yes Yes Short Spam

Yes No Long Not Spam

No Yes Short Spam

No No Long Not Spam

Yes Yes Long Spam

No No Short Not Spam

The new email has:

Offer = Yes

Free = No

Length = Short

We need to determine its class.

Step 1: Calculate Prior Probability

Prior probability means:

How likely is each class before looking at the features?

There are 6 emails.

Spam

There are 3 Spam emails.

Therefore:

$$\mathbf{P(\text{Spam}) = 3/6 = 0.5}$$

Not Spam

There are 3 Not Spam emails.

Therefore:

$$\mathbf{P(\text{Not Spam}) = 3/6 = 0.5}$$

So:

$$\mathbf{P(\text{Spam}) = 0.5}$$

$$\mathbf{P(\text{Not Spam}) = 0.5}$$

Step 2: Calculate Conditional Probabilities

Now we look at the features.

Offer

Among the 3 Spam emails:

- Offer = Yes $\rightarrow 2$
- Offer = No $\rightarrow 1$

Therefore:

$$\mathbf{P(\text{Offer=Yes} \mid \text{Spam}) = 2/3}$$

Among the 3 Not Spam emails:

- Offer = Yes $\rightarrow$ 1
- Offer = No $\rightarrow$ 2

Therefore:

$$\mathbf{P(\text{Offer=Yes} \mid \text{Not Spam}) = 1/3}$$

Free

Among Spam emails:

- Free = Yes $\rightarrow$ 2
- Free = No $\rightarrow$ 1

Therefore:

$$\mathbf{P(\text{Free=No} \mid \text{Spam}) = 1/3}$$

Among Not Spam emails:

- Free = Yes $\rightarrow$ 0
- Free = No $\rightarrow$ 3

Therefore:

$$\mathbf{P(\text{Free=No} \mid \text{Not Spam}) = 3/3 = 1}$$

Length

Among Spam emails:

- Short $\rightarrow$ 2
- Long $\rightarrow$ 1

Therefore:

$$\mathbf{P(\text{Short} \mid \text{Spam}) = 2/3}$$

Among Not Spam emails:

- Short $\rightarrow$ 1
- Long $\rightarrow$ 2

Therefore:

$$\mathbf{P(\text{Short} \mid \text{Not Spam}) = 1/3}$$

Step 3: Calculate Spam Probability

For the new email:

Offer = Yes, Free = No, Length = Short

Naïve Bayes calculates:

$$\mathbf{\text{Spam score} = P(\text{Spam}) \times P(\text{Offer}=\text{Yes}|\text{Spam}) \times P(\text{Free}=\text{No}|\text{Spam}) \times P(\text{Short}|\text{Spam})}$$

Substitute:

$$\mathbf{= 0.5 \times 2/3 \times 1/3 \times 2/3}$$

$$\mathbf{= 0.07407}$$

Step 4: Calculate Not Spam Probability

$$\mathbf{\text{Not Spam score} = P(\text{Not Spam}) \times P(\text{Offer}=\text{Yes}|\text{Not Spam}) \times P(\text{Free}=\text{No}|\text{Not Spam}) \times P(\text{Short}|\text{Not Spam})}$$

Substitute:

$$\mathbf{= 0.5 \times 1/3 \times 1 \times 1/3}$$

= 0.05556

Step 5: Compare

We have:

Spam = 0.07407

Not Spam = 0.05556

Since:

$$0.07407 > 0.05556$$

the model chooses:

SPAM

After normalization, approximately:

- **Spam = 57.14%**
- **Not Spam = 42.8%**

What is the Independence Assumption?

This is the final concept in Question 3.

Naïve Bayes assumes that the features are **independent of each other when the class is known**.

In our example, it assumes:

Offer, Free, and Length

contribute independently to deciding whether an email is Spam or Not Spam.

This assumption is not always perfectly true in real life, but it makes the calculation simple and fast.

